

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

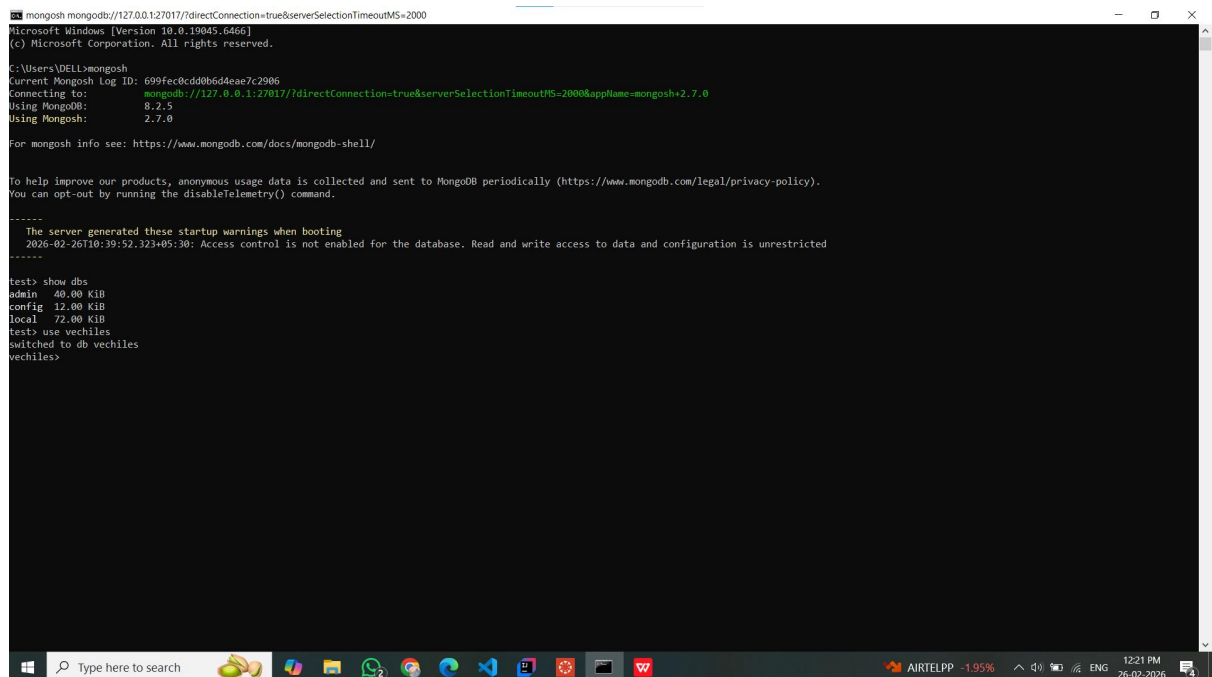
School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

1. Use MongoDB to implement the following DB operations

1. Create a database called 'vehicles' and *write* a MongoDB query to select database as "vehicles".



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>mongosh
Current Mongosh Log ID: 699fecdd8b6d4eae7c2906
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:  8.2.5
Using Mongosh:  2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

-----
The server generated these startup warnings when booting
2026-02-26T10:39:52.323+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin    40.00 KiB
config   12.00 KiB
local    72.00 KiB
test> use vehicles
switched to db vehicles
vehicles>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

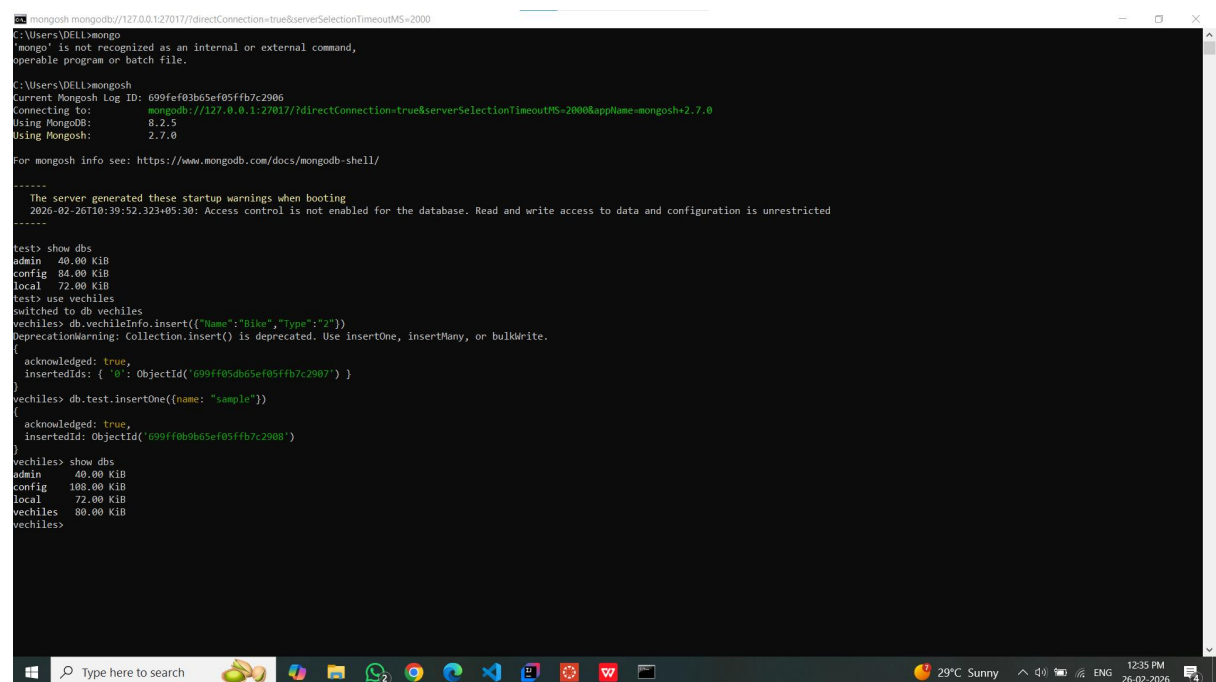
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2. Write a MongoDB query to display all the databases.



```

C:\Users\DELL>mongo
'mongo' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\DELL>mongosh
Current Mongosh Log ID: 699fef03b65ef05ffb7c2906
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:      8.2.5
Using Mongosh:      2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-02-26T10:39:52.323+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> show dbs
admin      40.00 KiB
config     84.00 KiB
local      72.00 KiB
test> use vechiles
switched to db vechiles
vechiles> db.vechileInfo.insert({"Name":"Bike","Type":"2"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699ff05db65ef05ffb7c2907') }
}
vechiles> db.test.insertOne({name: "sample"})
{
  acknowledged: true,
  insertedId: ObjectId('699ff0b9b65ef05ffb7c2908')
}
vechiles> show dbs
admin      40.00 KiB
config     108.00 KiB
local      72.00 KiB
vechiles   80.00 KiB
vechiles>
```

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Date: 26-02-2026

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3. Create a collection called 'two_wheelers'. (use capping) and Create a collection called 'four_wheelers'.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
test> show dbs
admin      40.00 KiB
config    84.00 KiB
local     72.00 KiB
test> use vechiles
switched to db vechiles
vechiles> db.vechilesInfo.insert({"Name":"Bike","Type":"2"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('699ff05db65ef05ffb7c2907') }
}
vechiles> db.test.insertOne({name: "sample"})
{
  acknowledged: true,
  insertedId: ObjectId('699ff0b9b65ef05ffb7c2908')
}
vechiles> show dbs
admin      40.00 KiB
config    108.00 KiB
local     72.00 KiB
vechiles  80.00 KiB
vechiles> db.createCollection("two_wheelers",{capped:true,size:5242880,max:100})
{ ok: 1 }
vechiles> db.createCollection("four_wheelers")
{ ok: 1 }
vechiles>
```

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4. Add 5 two-wheeler details to the collection named 'two_wheelers'. Each document consists of following fields as bike_name, model (gear or gearless), category (100cc, 125cc, 150cc, 200cc), colors_available (red, black, blue, sport red etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
vevachiles> db.two_wheelers.insertMany([
  {
    bike_name: "Shine 125",
    model: "gear",
    category: "125cc",
    colors_available: ["red", "black", "blue"],
    manufacturer: "Honda",
    performance: 8,
    timestamp: new Date("2022-03-15"),
    price: 85000
  },
  {
    bike_name: "Activa 66",
    model: "gearless",
    category: "100cc",
    colors_available: ["black", "grey", "white"],
    manufacturer: "Honda",
    performance: 9,
    timestamp: new Date("2023-01-10"),
    price: 78000
  },
  {
    bike_name: "Pulsar 150",
    model: "gear",
    category: "150cc",
    colors_available: ["black", "sport red"],
    manufacturer: "Bajaj",
    performance: 8,
    timestamp: new Date("2021-06-20"),
    price: 110000
  },
  {
    bike_name: "Apache RTR 160",
    model: "gear",
    category: "160cc",
    colors_available: ["red", "blue"],
    manufacturer: "TVS",
    performance: 9,
    timestamp: new Date("2022-09-05"),
    price: 120000
  },
  {
    bike_name: "FZ-S",
    model: "gear",
    category: "150cc",
    colors_available: ["black", "blue"],
    manufacturer: "Yamaha",
    performance: 7,
    timestamp: new Date("2020-11-11"),
    price: 115000
  }
])
```

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```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
{
  model: "gear",
  category: "150cc",
  colors_available: ["black", "sport red"],
  manufacturer: "Bajaj",
  performance: 8,
  timestamp: new Date("2021-06-20"),
  price: 110000
},
{
  bike_name: "Apache RTR 160",
  model: "gear",
  category: "150cc",
  colors_available: ["red", "blue"],
  manufacturer: "TVS",
  performance: 9,
  timestamp: new Date("2022-09-05"),
  price: 120000
},
{
  bike_name: "FZ-S",
  model: "gear",
  category: "150cc",
  colors_available: ["black", "blue"],
  manufacturer: "Yamaha",
  performance: 7,
  timestamp: new Date("2020-11-11"),
  price: 115000
}
}
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('699ff3d8b65ef05ffb7c2989'),
    '1': ObjectId('699ff3d8b65ef05ffb7c298a'),
    '2': ObjectId('699ff3d8b65ef05ffb7c298b'),
    '3': ObjectId('699ff3d8b65ef05ffb7c298c'),
    '4': ObjectId('699ff3d8b65ef05ffb7c298d')
  }
}
vehicles>
```

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Reg. no.: 23BCE9397

5. Add 5 four-wheeler details to the collection named 'four_wheelers'. Each document consists of following fields as vehicle_name, model (commercial or own), category (car, lorry, bus, mini truck, heavy truck, containers), variants (vxi, zxi, petrol, diesel etc) as array, manufacturer, performance (out of 10), timestamp (date and year release) and price.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
vechiles> db.four_wheelers.insertMany([
  {
    vehicle_name: "Ferrari 498 GTB",
    model: "own",
    category: "car",
    variants: ["petrol", "v8", "sports"],
    manufacturer: "Ferrari",
    performance: 10,
    timestamp: new Date("2022-06-15"),
    price: 45000000
  },
  {
    vehicle_name: "Bugatti Chiron",
    model: "own",
    category: "car",
    variants: ["petrol", "v16", "hyper sport"],
    manufacturer: "Bugatti",
    performance: 10,
    timestamp: new Date("2021-09-20"),
    price: 190000000
  },
  {
    vehicle_name: "McLaren 720S",
    model: "own",
    category: "car",
    variants: ["petrol", "v8", "coupe"],
    manufacturer: "McLaren",
    performance: 9,
    timestamp: new Date("2020-04-12"),
    price: 35000000
  },
  {
    vehicle_name: "Porsche 911 Turbo",
    model: "own",
    category: "car",
    variants: ["petrol", "turbo", "sports"],
    manufacturer: "Porsche",
    performance: 9,
    timestamp: new Date("2019-11-05"),
    price: 30000000
  }
])
vechiles>
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69a04f31d60d8c7d297c2987'),
    '1': ObjectId('69a04f31d60d8c7d297c2988'),
    '2': ObjectId('69a04f31d60d8c7d297c2989'),
    '3': ObjectId('69a04f31d60d8c7d297c298a')
  }
}
vechiles>
```

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6. Write a MongoDB query to display all documents available in two_wheelers and four_wheelers.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
teachiles> db.two_wheelers.find()
{
  "_id": ObjectId("699ff3d8b65ef05ffb7c2989"),
  "bike_name": "Shine 125",
  "model": "gear",
  "category": "125cc",
  "colors_available": [ "red", "black", "blue" ],
  "manufacturer": "Honda",
  "performance": 8,
  "timestamp": ISODate("2022-03-15T00:00:00.000Z"),
  "price": 85000
},
{
  "_id": ObjectId("699ff3d8b65ef05ffb7c298a"),
  "bike_name": "Activa 6G",
  "model": "gearless",
  "category": "110cc",
  "colors_available": [ "black", "grey", "white" ],
  "manufacturer": "Honda",
  "performance": 9,
  "timestamp": ISODate("2023-01-10T00:00:00.000Z"),
  "price": 78000
},
{
  "_id": ObjectId("699ff3d8b65ef05ffb7c298b"),
  "bike_name": "Pulsar 150",
  "model": "gear",
  "category": "150cc",
  "colors_available": [ "black", "sport red" ],
  "manufacturer": "Bajaj",
  "performance": 8,
  "timestamp": ISODate("2021-06-20T00:00:00.000Z"),
  "price": 110000
},
{
  "_id": ObjectId("699ff3d8b65ef05ffb7c298c"),
  "bike_name": "Apache RTR 160",
  "model": "gear",
  "category": "160cc",
  "colors_available": [ "red", "blue" ],
  "manufacturer": "TVS",
  "performance": 9,
  "timestamp": ISODate("2022-09-05T00:00:00.000Z"),
  "price": 120000
},
{
  "_id": ObjectId("699ff3d8b65ef05ffb7c298d"),
  "bike_name": "FZ-S",
  "model": "gear",
  "category": "160cc",
  "colors_available": [ "red", "blue" ],
  "manufacturer": "TVS",
  "performance": 9,
  "timestamp": ISODate("2022-09-05T00:00:00.000Z"),
  "price": 120000
}
```

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```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
{
  price: 78000
},
{
  _id: ObjectId('699ff3d8b65ef05ffb7c298b'),
  bike_name: 'Pulsar 150',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'black', 'sport red' ],
  manufacturer: 'Bajaj',
  performance: 8,
  timestamp: ISODate('2021-06-20T00:00:00.000Z'),
  price: 110000
},
{
  _id: ObjectId('699ff3d8b65ef05ffb7c298c'),
  bike_name: 'Apache RTR 160',
  model: 'gear',
  category: '160cc',
  colors_available: [ 'red', 'blue' ],
  manufacturer: 'TVS',
  performance: 9,
  timestamp: ISODate('2022-09-05T00:00:00.000Z'),
  price: 120000
},
{
  _id: ObjectId('699ff3d8b65ef05ffb7c298d'),
  bike_name: 'FZ-S',
  model: 'gear',
  category: '150cc',
  colors_available: [ 'black', 'blue' ],
  manufacturer: 'Yamaha',
  performance: 7,
  timestamp: ISODate('2020-11-11T00:00:00.000Z'),
  price: 115000
}
}
vechiles>
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
vechiles> db.four_wheelers.find()
{
  bike_name: 'Shine 125', price: 85000 },
  _id: ObjectId('69a04f31d60d8c7d297c2987'),
  vehicle_name: 'Ferrari 488 GTB', price: 1100000 },
  model: 'own', pache RTR 160', price: 120000 },
  category: 'car', price: 115000 },
  variants: [ 'petrol', 'v8', 'sports' ],
  manufacturer: 'Ferrari',
  performance: 10,
  timestamp: ISODate('2022-06-15T00:00:00.000Z'),
  price: 45000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c2988'),
  vehicle_name: 'Bugatti Chiron',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'v16', 'hyper sport' ],
  manufacturer: 'Bugatti',
  performance: 10,
  timestamp: ISODate('2021-09-20T00:00:00.000Z'),
  price: 190000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c2989'),
  vehicle_name: 'McLaren 720S',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'v8', 'coupe' ],
  manufacturer: 'McLaren',
  performance: 9,
  timestamp: ISODate('2020-04-12T00:00:00.000Z'),
  price: 35000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c298a'),
  vehicle_name: 'Porsche 911 Turbo',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'turbo', 'sports' ],
  manufacturer: 'Porsche',
  performance: 9,
  timestamp: ISODate('2019-11-05T00:00:00.000Z'),
  price: 30000000
}
}
vechiles>
```


Lab Sheet 6: MongoDB Basic commands

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7. Write a MongoDB query to display only vehicle name and price in all the collection of the database

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
performance: 10,
timestamp: ISODate('2022-06-15T00:00:00.000Z'),
price: 45000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c2988'),
  vehicle_name: 'Bugatti Chiron',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'w16', 'hyper sport' ],
  manufacturer: 'Bugatti',
  performance: 10,
  timestamp: ISODate('2021-09-20T00:00:00.000Z'),
  price: 190000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c2989'),
  vehicle_name: 'McLaren 720S',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'v8', 'coupe' ],
  manufacturer: 'McLaren',
  performance: 9,
  timestamp: ISODate('2020-04-12T00:00:00.000Z'),
  price: 35000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c298a'),
  vehicle_name: 'Porsche 911 Turbo',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'turbo', 'sports' ],
  manufacturer: 'Porsche',
  performance: 9,
  timestamp: ISODate('2019-11-05T00:00:00.000Z'),
  price: 30000000
}
}
vechiles> db.two_wheelers.find({}, {bike_name: 1, price: 1, _id: 0})
[
  { bike_name: 'Shine 125', price: 85000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Pulsar 150', price: 110000 },
  { bike_name: 'Apache RTR 160', price: 120000 },
  { bike_name: 'FZ-S', price: 115000 }
]
vechiles>
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
{
  _id: ObjectId('69a04f31d60d8c7d297c2989'),
  vehicle_name: 'McLaren 720S',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'v8', 'coupe' ],
  manufacturer: 'McLaren',
  performance: 9,
  timestamp: ISODate('2020-04-12T00:00:00.000Z'),
  price: 35000000
},
{
  _id: ObjectId('69a04f31d60d8c7d297c298a'),
  vehicle_name: 'Porsche 911 Turbo',
  model: 'own',
  category: 'car',
  variants: [ 'petrol', 'turbo', 'sports' ],
  manufacturer: 'Porsche',
  performance: 9,
  timestamp: ISODate('2019-11-05T00:00:00.000Z'),
  price: 30000000
}
}
vechiles> db.two_wheelers.find({}, {bike_name: 1, price: 1, _id: 0})
[
  { bike_name: 'Shine 125', price: 85000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Pulsar 150', price: 110000 },
  { bike_name: 'Apache RTR 160', price: 120000 },
  { bike_name: 'FZ-S', price: 115000 }
]
vechiles> db.four_wheelers.find({}, {vehicle_name: 1, price: 1, _id: 0})
[
  { price: 45000000 },
  { price: 190000000 },
  { price: 35000000 },
  { price: 30000000 }
]
vechiles>
```

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8. Write a MongoDB query to display two_wheelers from a particular company

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
>
vechiles> db.two_wheelers.find({}, {bike_name: 1, price: 1, _id: 0})
[
  { bike_name: 'Shine 125', price: 85000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Pulsar 150', price: 110000 },
  { bike_name: 'Apache RTR 160', price: 120000 },
  { bike_name: 'FZ-S', price: 115000 }
]
vechiles> db.four_wheelers.find({}, {vechile_name: 1, price: 1, _id: 0})
[
  { price: 45000000 },
  { price: 190000000 },
  { price: 350000000 },
  { price: 300000000 }
]
vechiles> db.two_wheelers.find({manufacturer: "Honda"})
[
  {
    _id: ObjectId('699ff3d8b65ef05ffb7c2909'),
    bike_name: 'Shine 125',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-03-15T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('699ff3d8b65ef05ffb7c290a'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'black', 'grey', 'white' ],
    manufacturer: 'Honda',
    performance: 9,
    timestamp: ISODate('2023-01-10T00:00:00.000Z'),
    price: 78000
  }
]
vechiles>
```

9. Write a MongoDB query to display four_wheelers available in diesel variants

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
>
vechiles> db.four_wheelers.find({}, {vechile_name: 1, price: 1, _id: 0})
[
  { bike_name: 'Shine 125', price: 85000 },
  { bike_name: 'Activa 6G', price: 78000 },
  { bike_name: 'Pulsar 150', price: 110000 },
  { bike_name: 'Apache RTR 160', price: 120000 },
  { bike_name: 'FZ-S', price: 115000 }
]
vechiles> db.four_wheelers.find({manufacturer: "Honda"})
[
  {
    _id: ObjectId('699ff3d8b65ef05ffb7c2909'),
    bike_name: 'Shine 125',
    model: 'gear',
    category: '125cc',
    colors_available: [ 'red', 'black', 'blue' ],
    manufacturer: 'Honda',
    performance: 8,
    timestamp: ISODate('2022-03-15T00:00:00.000Z'),
    price: 85000
  },
  {
    _id: ObjectId('699ff3d8b65ef05ffb7c290a'),
    bike_name: 'Activa 6G',
    model: 'gearless',
    category: '110cc',
    colors_available: [ 'black', 'grey', 'white' ],
    manufacturer: 'Honda',
    performance: 9,
    timestamp: ISODate('2023-01-10T00:00:00.000Z'),
    price: 78000
  }
]
vechiles>
```

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Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

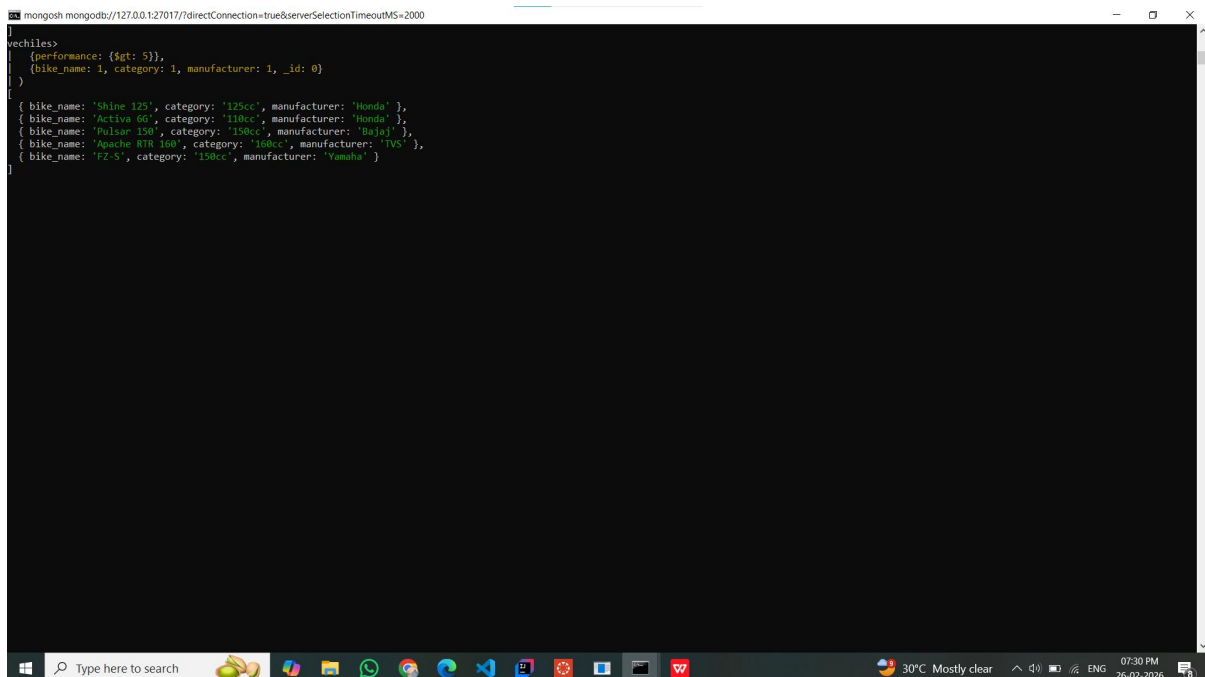
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10. Write a MongoDB query to display vehicles name, category and manufacturer details whose rating is more than 5.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
> use vehicles
> {performance: {$gt: 5}},
> {bike_name: 1, category: 1, manufacturer: 1, _id: 0}
>
[
  { bike_name: 'Shine 125', category: '125cc', manufacturer: 'Honda' },
  { bike_name: 'Activa 6G', category: '110cc', manufacturer: 'Honda' },
  { bike_name: 'Pulsar 150', category: '150cc', manufacturer: 'Bajaj' },
  { bike_name: 'Apache RTR 160', category: '160cc', manufacturer: 'TVS' },
  { bike_name: 'FZ-S', category: '130cc', manufacturer: 'Yamaha' }
]
```

The screenshot shows a MongoDB command prompt window. The user has entered the command `use vehicles` and then a query `{performance: {$gt: 5}}, {bike_name: 1, category: 1, manufacturer: 1, _id: 0}`. The results are displayed as a JSON array of objects, each representing a vehicle with its name, category, and manufacturer. The results are: `[{ bike_name: 'Shine 125', category: '125cc', manufacturer: 'Honda' }, { bike_name: 'Activa 6G', category: '110cc', manufacturer: 'Honda' }, { bike_name: 'Pulsar 150', category: '150cc', manufacturer: 'Bajaj' }, { bike_name: 'Apache RTR 160', category: '160cc', manufacturer: 'TVS' }, { bike_name: 'FZ-S', category: '130cc', manufacturer: 'Yamaha' }]`. The window title is `mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000`. The Windows taskbar is visible at the bottom, showing the time as 07:30 PM on 26-02-2026.

VIT-AP UNIVERSITY, ANDHRA PRADESH

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
{
  manufacturer: 'Honda',
  performance: 9,
  timestamp: ISODate('2023-01-10T00:00:00.000Z'),
  price: 78000
}
}
vechiles> db.four-wheelers.find(
  {
    performance: { $gt: 5 },
    { vehicle_name: 1, category: 1, manufacturer: 1, _id: 0 }
  }
)
[
  {
    bike_name: 'Shine 125', category: '125cc', manufacturer: 'Honda' },
    vehicle_name: 'Ferrari 488 GTB', category: '110cc', manufacturer: 'Honda' },
    category: 'car', bike_name: 'Bajaj 150', category: '150cc', manufacturer: 'Bajaj' },
    manufacturer: 'Ferrari 160', category: '160cc', manufacturer: 'TVS' },
    bike_name: 'FZ-S', category: '150cc', manufacturer: 'Yamaha' }
  ],
  {
    vehicle_name: 'Bugatti Chiron',
    category: 'car',
    manufacturer: 'Bugatti'
  },
  {
    vehicle_name: 'McLaren 720S',
    category: 'car',
    manufacturer: 'McLaren'
  },
  {
    vehicle_name: 'Porsche 911 Turbo',
    category: 'car',
    manufacturer: 'Porsche'
  }
]
vechiles>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

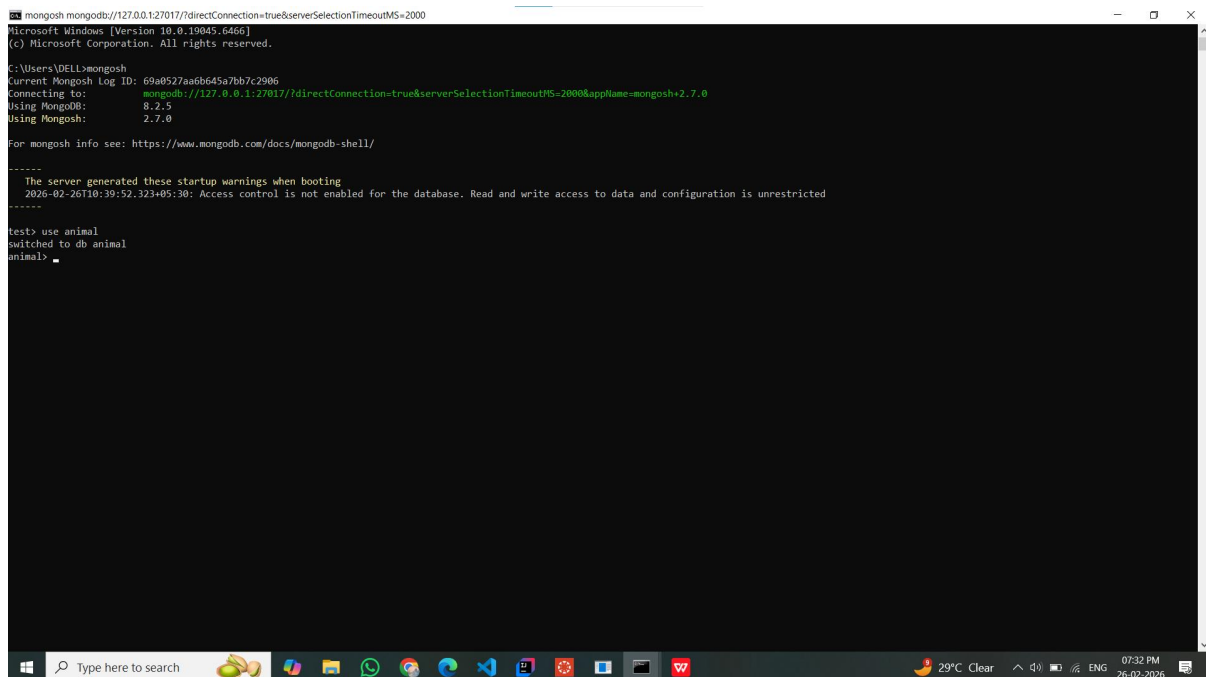
School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

2. Use MongoDB to implement the following DB operations for a Zoo

1. Create a database called 'animal' and *write* a MongoDB query to select database as 'animal'.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>mongosh
Current Mongosh Log ID: 69a8527aa6b645a7bb7c2906
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:  8.2.5
Using Mongosh:  2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-02-26T10:39:52.323+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> use animal
switched to db animal
animal>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

2. Write a MongoDB query to display all the databases.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.19045.6466]
(c) Microsoft Corporation. All rights reserved.

C:\Users\DELL>mongosh
Current Mongosh Log ID: 69a8527a6b645a7bb7c2906
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.7.0
Using MongoDB:  8.2.5
Using Mongosh:  2.7.0

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
  The server generated these startup warnings when booting
  2026-02-26T10:39:52.323+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
  -----

test> use animal
switched to db animal
animal> db.insertOne()
TypeError: db.insertOne is not a function
animal> db.insertOne({"Animal":"lion"})
TypeError: db.insertOne is not a function
animal> db.insertOne();
TypeError: db.insertOne is not a function
animal> db.insertOne({"Animal":"lion"});
TypeError: db.insertOne is not a function
animal> db.Animal.insertOne({"Animal": "lion"});
{
  acknowledged: true,
  insertedId: ObjectId('69a852d4fab645a7bb7c2907')
}
animal> show dbs
admin      40.00 KiB
animal     8.00 KiB
config    104.00 KiB
local     72.00 KiB
replsets  160.00 KiB
animal>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

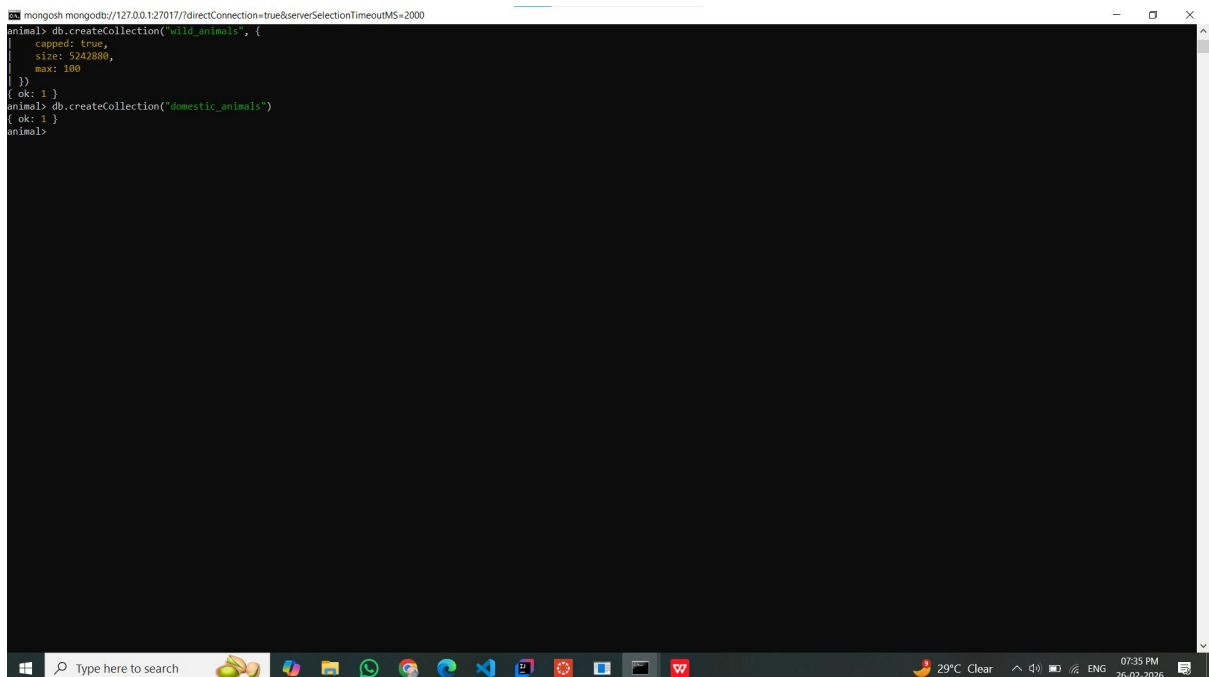
Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

3. Create a collection called 'wild_animals'.(use capping) and Create a collection called 'domestic_animals'.



```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.createCollection('wild_animals', {
  capped: true,
  size: 5242880,
  max: 100
})
{ ok: 1 }
animal> db.createCollection("domestic_animals")
{ ok: 1 }
animal>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

4. Add 5 wild_animal details to the collection named 'wild_animals'. Each document consists of following fields as animal_name, nature (harm or harmless), favorite_foods (meat, rabbits, deer etc) as array, care_taker_name, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.wild_animals.insertMany([
  {
    animal_name: "Lion",
    nature: "harm",
    favorite_foods: ["meat", "deer"],
    care_taker_name: "Ravi",
    life_span: 14,
    timestamp: new Date("2022-01-15"),
    expenses: 50000
  },
  {
    animal_name: "Tiger",
    nature: "harm",
    favorite_foods: ["meat", "rabbits"],
    care_taker_name: "Suresh",
    life_span: 16,
    timestamp: new Date("2021-06-10"),
    expenses: 60000
  },
  {
    animal_name: "Elephant",
    nature: "harmless",
    favorite_foods: ["grass", "fruits"],
    care_taker_name: "Manoj",
    life_span: 60,
    timestamp: new Date("2020-03-20"),
    expenses: 80000
  },
  {
    animal_name: "Deer",
    nature: "harmless",
    favorite_foods: ["grass"],
    care_taker_name: "Ravi",
    life_span: 20,
    timestamp: new Date("2023-02-11"),
    expenses: 20000
  },
  {
    animal_name: "Bear",
    nature: "harm",
    favorite_foods: ["fish", "meat"],
    care_taker_name: "Kiran",
    life_span: 25,
    timestamp: new Date("2019-11-05"),
    expenses: 45000
  }
])
```


Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
{
  animal_name: "Deer",
  nature: "harmless",
  favorite_foods: ["grass"],
  care_taker_name: "Ravi",
  life_span: 20,
  timestamp: new Date("2023-02-11"),
  expenses: 20000
},
{
  animal_name: "Bear",
  nature: "harm",
  favorite_foods: ["fish", "meat"],
  care_taker_name: "Kiran",
  life_span: 25,
  timestamp: new Date("2019-11-05"),
  expenses: 45000
}
}
(
acknowledged: true,
insertedIds: {
  '0': ObjectId('69a0534ca6b645a7bb7c2908'),
  '1': ObjectId('69a0534ca6b645a7bb7c2909'),
  '2': ObjectId('69a0534ca6b645a7bb7c290a'),
  '3': ObjectId('69a0534ca6b645a7bb7c290b'),
  '4': ObjectId('69a0534ca6b645a7bb7c290c')
}
)
animal> _
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

5. Add 5 domestic-animal details to the collection named 'domestic_animals'. Each document consists of following fields as animal_name, gender (male or female), favorite_foods (meat, rabbits, deer etc) as array, animal_petname, life span (in years), timestamp (when the animal registered at the Zoo) and expenses.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.domestic_animals.insertMany([
  {
    animal_name: "Cow",
    gender: "female",
    favorite_foods: ["grass", "fodder"],
    animal_petname: "Lakshmi",
    life_span: 18,
    timestamp: new Date("2021-05-14"),
    expenses: 15000
  },
  {
    animal_name: "Dog",
    gender: "male",
    favorite_foods: ["meat", "pedigree"],
    animal_petname: "Tomy",
    life_span: 12,
    timestamp: new Date("2022-07-19"),
    expenses: 10000
  },
  {
    animal_name: "Cat",
    gender: "female",
    favorite_foods: ["fish", "milk"],
    animal_petname: "Kitty",
    life_span: 15,
    timestamp: new Date("2023-01-10"),
    expenses: 8000
  },
  {
    animal_name: "Goat",
    gender: "male",
    favorite_foods: ["grass"],
    animal_petname: "Maju",
    life_span: 14,
    timestamp: new Date("2020-09-22"),
    expenses: 9000
  },
  {
    animal_name: "Horse",
    gender: "female",
    favorite_foods: ["hay", "grains"],
    animal_petname: "Star",
    life_span: 25,
    timestamp: new Date("2018-12-30"),
    expenses: 30000
  }
])
```


Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

6. Write a MongoDB query to display all documents available in wild_animals and domestic_animals.

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animals> db.wild_animals.find()
[
  {
    _id: ObjectId('69a0534ca6b645a7bb7c2908'),
    animal_name: 'Lion',
    nature: 'harm',
    favorite_foods: [ 'meat', 'deer' ],
    care_taker_name: 'Ravi',
    life_span: 14,
    timestamp: ISODate('2022-01-15T00:00:00.000Z'),
    expenses: 50000
  },
  {
    _id: ObjectId('69a0534ca6b645a7bb7c2909'),
    animal_name: 'Tiger',
    nature: 'harm',
    favorite_foods: [ 'meat', 'rabbits' ],
    care_taker_name: 'Suresh',
    life_span: 16,
    timestamp: ISODate('2021-06-10T00:00:00.000Z'),
    expenses: 60000
  },
  {
    _id: ObjectId('69a0534ca6b645a7bb7c290a'),
    animal_name: 'Elephant',
    nature: 'harmless',
    favorite_foods: [ 'grass', 'fruits' ],
    care_taker_name: 'Manoj',
    life_span: 66,
    timestamp: ISODate('2020-03-20T00:00:00.000Z'),
    expenses: 80000
  },
  {
    _id: ObjectId('69a0534ca6b645a7bb7c290b'),
    animal_name: 'Deer',
    nature: 'harmless',
    favorite_foods: [ 'grass' ],
    care_taker_name: 'Ravi',
    life_span: 20,
    timestamp: ISODate('2023-02-11T00:00:00.000Z'),
    expenses: 20000
  },
  {
    _id: ObjectId('69a0534ca6b645a7bb7c290c'),
    animal_name: 'Bear',
    nature: 'harm',
    favorite_foods: [ 'fish', 'meat' ],
    care_taker_name: 'Kiran',
    life_span: 25,
    timestamp: ISODate('2019-11-05T00:00:00.000Z'),
    expenses: 30000
  }
]
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.domestic_animals.find()
[
  {
    _id: ObjectId('69a0537fa6b645a7bb7c290d'),
    animal_name: 'Cow',
    gender: 'female',
    favorite_foods: [ 'grass', 'fodder' ],
    animal_petname: 'lakshmi',
    life_span: 18,
    timestamp: ISODate('2021-05-14T00:00:00.000Z'),
    expenses: 15000
  },
  {
    _id: ObjectId('69a0537fa6b645a7bb7c290e'),
    animal_name: 'Dog',
    gender: 'male',
    favorite_foods: [ 'meat', 'pedigree' ],
    animal_petname: 'Tommy',
    life_span: 12,
    timestamp: ISODate('2022-07-19T00:00:00.000Z'),
    expenses: 10000
  },
  {
    _id: ObjectId('69a0537fa6b645a7bb7c290f'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: ISODate('2023-01-10T00:00:00.000Z'),
    expenses: 8000
  },
  {
    _id: ObjectId('69a0537fa6b645a7bb7c2910'),
    animal_name: 'Goat',
    gender: 'male',
    favorite_foods: [ 'grass' ],
    animal_petname: 'Raju',
    life_span: 14,
    timestamp: ISODate('2020-09-22T00:00:00.000Z'),
    expenses: 9000
  },
  {
    _id: ObjectId('69a0537fa6b645a7bb7c2911'),
    animal_name: 'Horse',
    gender: 'female',
    favorite_foods: [ 'hay', 'grains' ],
    animal_petname: 'Star',
    life_span: 25,
    timestamp: ISODate('2018-12-30T00:00:00.000Z')
  }
]
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

7. Write a MongoDB query to display only animal name and expenses in all the collection of the database

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.wild_animals.find({}, {animal_name: 1, expenses: 1, _id: 0})
[
  { animal_name: 'Lion', expenses: 50000 },
  { animal_name: 'Tiger', expenses: 60000 },
  { animal_name: 'Elephant', expenses: 80000 },
  { animal_name: 'Deer', expenses: 20000 },
  { animal_name: 'Bear', expenses: 45000 }
]
animal> db.domestic_animals.find({}, {animal_name: 1, expenses: 1, _id: 0})
[
  { animal_name: 'Cow', expenses: 15000 },
  { animal_name: 'Dog', expenses: 10000 },
  { animal_name: 'Cat', expenses: 8000 },
  { animal_name: 'Goat', expenses: 9000 },
  { animal_name: 'Horse', expenses: 30000 }
]
animal>
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

8. Write a MongoDB query to display domestic_animals whose life is a particular year

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
animal> db.domestic_animals.find({life_span: 15})
[
  {
    _id: ObjectId('69a0537fa0b645a7bb7c290f'),
    animal_name: 'Cat',
    gender: 'female',
    favorite_foods: [ 'fish', 'milk' ],
    animal_petname: 'Kitty',
    life_span: 15,
    timestamp: ISODate('2023-01-10T00:00:00.000Z'),
    expenses: 8000
  }
]
```

Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

Student name: Mahammad Galeeb

Reg. no.: 23BCE9397

9. Write a MongoDB query to display wild_animals available under a particular care_taker

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
> use animals
> db.wild_animals.find({care_taker_name: "Ravi"})
{
  "_id": ObjectId("69a0534ca0b645a7bb7c2908"),
  "animal_name": "Lion",
  "nature": "Rare",
  "favorite_foods": [ "meat", "deer" ],
  "care_taker_name": "Ravi",
  "life_span": 14,
  "timestamp": ISODate("2022-01-15T00:00:00.000Z"),
  "expenses": 50000
},
{
  "_id": ObjectId("69a0534ca0b645a7bb7c290b"),
  "animal_name": "Deer",
  "nature": "Harmless",
  "favorite_foods": [ "grass" ],
  "care_taker_name": "Ravi",
  "life_span": 20,
  "timestamp": ISODate("2023-02-11T00:00:00.000Z"),
  "expenses": 20000
}
>
```


Lab Sheet 6: MongoDB Basic commands

Branch/ Class: B.Tech

Date: 26-02-2026

Faculty Name: Prof. S.Gopikrishnan

School: SCOPE

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Reg. no.: 23BCE9397

10. Write a MongoDB query to display animal name, favorite_foods and expenses details whose lifespan is more than 5 years.

```
mongosh mongodb://127.0.0.1:27017/7?directConnection=true&serverSelectionTimeoutMS=2000
> use animal
> db.wild_animals.find(
  {
    (life_span: {$gt: 5}),
    {animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0}
  }
)
{
  animal_name: 'Lion',
  favorite_foods: [ 'meat', 'deer' ],
  expenses: 50000
},
{
  animal_name: 'Tiger',
  favorite_foods: [ 'meat', 'rabbits' ],
  expenses: 60000
},
{
  animal_name: 'Elephant',
  favorite_foods: [ 'grass', 'fruits' ],
  expenses: 80000
},
{
  animal_name: 'Deer', favorite_foods: [ 'grass' ], expenses: 20000 },
{
  animal_name: 'Bear',
  favorite_foods: [ 'fish', 'meat' ],
  expenses: 45000
}
> db.domestic_animals.find(
  {
    (life_span: {$gt: 5}),
    {animal_name: 1, favorite_foods: 1, expenses: 1, _id: 0}
  }
)
{
  animal_name: 'Cow',
  favorite_foods: [ 'grass', 'fodder' ],
  expenses: 15000
},
{
  animal_name: 'Dog',
  favorite_foods: [ 'meat', 'pedigree' ],
  expenses: 10000
},
{
  animal_name: 'Cat',
  favorite_foods: [ 'fish', 'milk' ],
  expenses: 8000
},
{
  animal_name: 'Goat', favorite_foods: [ 'grass' ], expenses: 9000 },
{
  animal_name: 'Horse',
```